

Module 4

Data-Only Full-Field Neural Operator Baseline

A textbook-style introduction to supervised operator learning for reduced resistive MHD

Learning task

Learn the time-conditioned solution operator

$$(\psi_0(x, y), U_0(x, y), \eta, \nu, t) \mapsto (\hat{\psi}(x, y, t), \hat{U}(x, y, t))$$

from verified full-field numerical arrays alone. No reduced-MHD equation residual, initial-condition penalty, boundary-condition penalty, gauge penalty, or global-balance penalty may update the Module 4 network.

Preface: why the baseline changed from a coordinate network to a full-field operator

Module 1 fixed the reduced resistive-MHD physics. Module 2 built and verified a numerical solver that produces complete two-dimensional trajectories. Module 3 now exposes those trajectories as lossless full-field operator examples. The natural data-only learning problem has therefore changed. The earlier Module 4 formulation asked a coordinate network to learn

$$(x, y, t, \eta, \nu, A_0) \mapsto (\psi, U).$$

That formulation represents a field by querying one point at a time. The revised formulation instead supplies the complete initial magnetic-flux and stream-function arrays and requests the complete target arrays at a chosen time:

$$(\psi_0, U_0, \eta, \nu, t) \mapsto (\psi_t, U_t).$$

This is an *operator-learning* problem because the input and output contain functions represented numerically over the spatial domain, not merely a few scalar coordinates.

The revision is scientifically useful for two reasons. First, simulation codes naturally produce fields. A full-field surrogate therefore resembles the way a later high-fidelity simulation export would actually be consumed. Second, supplying the complete initial condition permits future datasets to vary the *shape* of the initial plasma state, rather than describing every state through a small collection of hand-selected scalar amplitudes.

The revision does *not* change the physical model, the reference solver, or the scientific claim boundary. The training arrays are still synthetic outputs of the project’s two-dimensional reduced-MHD solver. They are not genuine M3D-C1 arrays. Likewise, the full-field tensors are numerical scalar arrays, not pictures.

Important distinction: numerical fields are not images

The neural operator consumes the stored scalar values of ψ_0 , U_0 , ψ_t , and U_t on the declared grid. Contour plots, magnetic-field-line renderings, velocity arrows, PNG files, screenshots, color maps, and rasterized figures are presentation products and may not replace the numerical arrays in training.

Important distinction: Module 4 remains data-only

A neural operator is not automatically physics-informed. Module 4 uses only labeled input-target field pairs to update its weights. Reduced-MHD equations may be evaluated after training as diagnostics, but if a PDE residual or other physical constraint contributes to the training objective, the experiment belongs to Module 5.

Important distinction: claim boundary

Completing Module 4 demonstrates a data-only full-field surrogate of the project’s synthetic two-field reduced resistive-MHD solution family. It does not demonstrate fidelity to genuine M3D-C1 output, realistic fusion geometry, extended-MHD closures, or arbitrary plasma initial conditions.

How to read this note

The main chain is

accepted trajectories $\longrightarrow$ full-field operator pairs $\longrightarrow$ whole-case partitions
 $\longrightarrow$ training-only transforms $\longrightarrow$ Fourier neural operator
 $\longrightarrow$ strict field-data loss $\longrightarrow$ optimization $\longrightarrow$ complete-field evaluation.

Sections 1–3 define the scientific task. Sections 4–6 build the full-field representation and the Fourier neural operator. Sections 10–11 define supervised training. Sections 14–19 define evaluation and the fair comparison with Module 5. Sections 20–25 freeze the version-1 baseline.

Learning objectives

After working through this module, the reader should be able to:

1. explain why Module 4 is still required when the project’s final interest is physics-informed learning;
2. distinguish a coordinate network, a neural time-stepper, and a time-conditioned neural operator;
3. write the exact full-field map $(\psi_0, U_0, \eta, \nu, t) \mapsto (\hat{\psi}_t, \hat{U}_t)$;
4. explain why A_0 remains metadata but is not duplicated as a primary operator input in version 1;
5. distinguish a physical trajectory from the many target-time operator pairs extracted from it;
6. explain why whole physical trajectories must be split before target-time pair extraction;
7. distinguish raw numerical fields from rendered images;
8. describe tensor channel order, grid order, and reversible preprocessing;
9. explain what a neural operator learns conceptually;
10. derive the main steps of a two-dimensional Fourier neural operator layer;

11. explain spectral convolution, retained modes, local channel mixing, lifting, and projection;
12. explain how scalar time and transport coefficients condition a full-field operator;
13. construct a channel-balanced and case-balanced full-field data loss;
14. explain mini-batch training, backpropagation, optimization, and checkpoint selection for field tensors;
15. evaluate complete unseen fields using ψ, U, J, ω , energies, reconnection diagnostics, and evaluation-only PDE residuals;
16. distinguish parameter interpolation, time interpolation, initial-condition generalization, and resolution transfer;
17. identify the present limitation that $U_0 = A_0(\cos x - \cos y)$ spans only an amplitude family;
18. design a controlled Module 4 versus Module 5 comparison; and
19. state precisely what a successful result does and does not establish about M3D-C1.

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1 Scientific role of Module 4

1.1 Four computational objects

Several project components produce field values, but their meanings differ.

Object	What it does	What determines its output
Reduced-MHD reference solver	Numerically integrates the frozen two-field equations from one initial condition.	PDEs, initial fields, parameters, grid, time step, dealiasing, integrator, and numerical conventions.
Canonical trajectory dataset	Stores accepted numerical arrays and provenance.	Verified Module 2 runs and the Module 3 schema.
Module 4 neural operator	Learns complete target fields from complete initial fields and scalar context using labeled data only.	Operator architecture, labeled field pairs, transforms, loss, sampling, optimizer, and training choices.
Module 5 physics-informed operator	Uses the same operator task but additionally penalizes declared reduced-MHD defects or constraints.	Module 4 ingredients plus physics information and its optimization strategy.

1.2 The controlled scientific question

Let $\mathcal{G}_{\text{data}}$ denote the trained data-only operator and $\mathcal{G}_{\text{phys}}$ the later physics-informed operator. The central question is not merely whether $\mathcal{G}_{\text{phys}}$ is accurate. It is

At the same labeled full-field data budget and on the same untouched physical cases, what changes when reduced-MHD information is added to a competent full-field operator baseline?

Measured outcomes can include field error, derivative error, current-sheet fidelity, reconnection timing, data efficiency, parameter generalization, initial-state generalization, runtime, memory, and optimization reliability.

A negative result is scientifically valid. A physics-informed operator may optimize less reliably, may be sensitive to residual scaling, or may add little when dense field observations already determine the solution family.

1.3 What “baseline” means

Definition

A **baseline** is a fully specified comparison method that answers the same prediction question under controlled conditions. A baseline should be competent rather than intentionally weak. Otherwise improvement can be attributed to avoidable defects in the comparator instead of the new method.

For the present periodic structured-grid problem, a competent baseline should use the correct tensor contract, training-only normalization, whole-case splits, adequate spectral resolution in the network, stable optimization, and complete-field evaluation.

1.4 Why the full-field baseline is scientifically different from the old coordinate baseline

A coordinate network learns field values through pointwise queries. A full-field operator instead learns a map between discretized functions. The change affects what constitutes one example, how batches are formed, what architectural inductive bias is useful, what generalization means, and how inference cost is measured.

It does *not* change the underlying target trajectory. For every accepted case c and time t_n , both formulations attempt to approximate the same numerical fields $\psi^{(c)}(x, y, t_n)$ and $U^{(c)}(x, y, t_n)$.

2 From trajectories to supervised operator learning

2.1 Supervised learning with field-valued inputs and outputs

In ordinary supervised learning an example is an input paired with a desired output. Here both contain arrays. Let

$$X_{c,n} = \left(\psi_0^{(c)}, U_0^{(c)}, \eta_c, \nu_c, t_n \right), \quad Y_{c,n} = \left(\psi_n^{(c)}, U_n^{(c)} \right).$$

A neural operator with trainable parameters θ predicts

$$\hat{Y}_{c,n} = \mathcal{G}_\theta(X_{c,n}).$$

Training changes θ to reduce disagreement between complete predicted and target fields.

2.2 One trajectory produces many pairs but not many independent cases

Suppose one accepted case contains N_t saved times. The same initial state may be paired with several target times,

$$(\psi_0, U_0, \eta, \nu, t_1) \mapsto (\psi_1, U_1), \quad \dots, \quad (\psi_0, U_0, \eta, \nu, t_{N_t}) \mapsto (\psi_{N_t}, U_{N_t}).$$

These are distinct supervised examples but share the same physical trajectory. If 27 trajectories each contribute 100 target times, the dataset contains 2700 target-time pairs but only 27 independent physical cases.

Important distinction: pair-count inflation

Do not describe thousands of target-time pairs as thousands of independent simulations. Temporal snapshots from one trajectory are correlated and inherit the same initial condition and transport parameters.

2.3 The $t = 0$ pair

At $t = 0$ the exact operator target is the input state itself:

$$\mathcal{S}_0(\psi_0, U_0, \eta, \nu) = (\psi_0, U_0).$$

This identity pair is useful as an acceptance test for tensor packing, transforms, channel order, and operator behavior. It should not dominate the training distribution merely because it is easy.

2.4 Approximation, interpolation, extrapolation, and generalization

The word *generalization* is incomplete unless the held-out dimension is stated. Relevant tests include:

- **transport-parameter interpolation:** unseen (η, ν) inside a declared training support;
- **transport-parameter extrapolation:** (η, ν) outside that support;
- **time interpolation:** target times inside the represented training interval;
- **time extrapolation:** $t > T_{\text{train}}$;
- **initial-state interpolation/generalization:** unseen initial fields from a family represented during training;
- **out-of-family initial states:** new functional shapes not represented in training; and
- **resolution transfer:** evaluation on a different spatial grid.

These tests answer different scientific questions and must not be averaged into one unnamed “generalization” score.

3 The exact Module 4 operator map

3.1 Primary map

For case c and target time t ,

$$\mathcal{G}_\theta : \left(\psi_0^{(c)}(x, y), U_0^{(c)}(x, y), \eta_c, \nu_c, t \right) \mapsto \left(\hat{\psi}^{(c)}(x, y, t), \hat{U}^{(c)}(x, y, t) \right). \quad (3.1)$$

All physical quantities are dimensionless under Module 1 normalization before optional neural preprocessing.

The operator is **time-conditioned**: it predicts a requested state directly from the initial state and target time. It is not required to predict intermediate snapshots sequentially.

3.2 Why A_0 is not duplicated as a primary input

The version-1 initial stream function is

$$U_0(x, y) = A_0(\cos x - \cos y).$$

Therefore the entire U_0 array already contains the perturbation amplitude. Supplying both U_0 and A_0 would duplicate the same information for the current family. Module 3 retains A_0 as metadata and for consistency checks, but the primary operator context is

$$(\eta, \nu, t),$$

not (η, ν, A_0, t) .

Important distinction: current initial-condition family is narrow

Replacing the scalar A_0 by the complete U_0 field does not magically create initial-condition diversity. Version 1 still uses the fixed shape $\cos x - \cos y$ multiplied by different amplitudes. A strong claim of operator generalization to arbitrary plasma states requires additional independently verified initial-field families.

3.3 Why J and ω remain derived outputs

The primary target remains (ψ, U) . Current and vorticity are derived by

$$J = -\nabla_{\perp}^2 \psi, \quad \omega = \nabla_{\perp}^2 U.$$

Predicting J and ω as independent target channels would answer a different learning question and can permit inconsistent fields. They are valuable evaluation diagnostics because they test whether the operator reproduces spatial curvature.

3.4 Coordinate network, time-stepper, and neural operator

Formulation	Typical map	Interpretation
Coordinate network	$(x, y, t, \mu) \mapsto q$	Returns one field value at a requested coordinate.
Autoregressive time-stepper	$q(t_n) \mapsto q(t_{n+1})$	Advances a state repeatedly; errors can accumulate recursively.
Time-conditioned neural operator	$(q_0, \mu, t) \mapsto q(t)$	Maps a complete initial function and scalar context directly to a complete target function.

Module 4 uses the third formulation.

3.5 What the operator is approximating

For the frozen reduced-MHD initial-value problem there is conceptually a solution operator

$$\mathcal{S}_t : (\psi_0, U_0, \eta, \nu) \mapsto (\psi_t, U_t).$$

The data-only model seeks

$$\mathcal{G}_{\theta} \approx \mathcal{S}_t$$

over the declared domain of initial fields, transport coefficients, and times. Because the targets come from a finite numerical solver, the more precise statement is that $\mathcal{G}_{\theta}$ approximates the *accepted discrete reference solution operator* represented by Module 2 and Module 3.

4 Full-field tensors and scientific data semantics

4.1 Canonical arrays versus learning tensors

The canonical HDF5 trajectory is the authoritative scientific object. The learning tensor is a reversible view. For a common $N_x \times N_y$ grid, Module 3 defines

$$\mathbf{X} \in \mathbb{R}^{B \times N_x \times N_y \times C_{\text{in}}}, \quad \mathbf{Y} \in \mathbb{R}^{B \times N_x \times N_y \times 2},$$

for batch size B , before any framework-specific channel-first permutation.

4.2 Version-1 input channels

The frozen primary baseline broadcasts scalar context as constant spatial channels. The channel-last input is

$$X[:, :, 0] = \tilde{\psi}_0(x, y), \tag{4.1}$$

$$X[:, :, 1] = \tilde{U}_0(x, y), \tag{4.2}$$

$$X[:, :, 2] = \tilde{\eta}, \tag{4.3}$$

$$X[:, :, 3] = \tilde{\nu}, \tag{4.4}$$

$$X[:, :, 4] = \tilde{t}. \tag{4.5}$$

Thus $C_{\text{in}} = 5$. The last three arrays are constant over (x, y) for one operator example.

Broadcasting does not claim that η , ν , or t are spatially varying. It is simply an architectural representation that places scalar context in the same tensor consumed by the lifting layer.

4.3 Target channels

The output channels are

$$Y[:, :, 0] = \tilde{\psi}_t(x, y), \quad Y[:, :, 1] = \tilde{U}_t(x, y).$$

Inverse transforms recover physical nondimensional fields before scientific evaluation.

4.4 Logical axis order versus framework order

A logical tensor may be stored as $[B, N_x, N_y, C]$ and then permuted to $[B, C, N_x, N_y]$ for a deep-learning framework. These are memory-layout choices, not changes in physics. Every permutation must be explicit and reversible.

4.5 Why random crops are not part of the primary baseline

Computer-vision pipelines often crop images. That would alter the physical periodic domain and discard global topology. The version-1 operator uses the full $2\pi \times 2\pi$ field. Patch-based or domain-decomposed operators are separate experiments requiring boundary and overlap definitions.

4.6 Common-grid requirement

The primary FNO batch assumes a common periodic grid. One accepted production realization per physical case contributes to the learning release with identical N_x, N_y, L_x, L_y and coordinate arrays. Verification runs at other resolutions remain linked provenance rather than silently resampled training cases.

5 What a neural operator means

5.1 From finite-dimensional vectors to functions

A standard network maps vectors to vectors,

$$\mathbb{R}^{d_{\text{in}}} \rightarrow \mathbb{R}^{d_{\text{out}}}.$$

An operator conceptually maps functions to functions,

$$\mathcal{G} : \mathcal{A} \rightarrow \mathcal{U},$$

where $\mathcal{A}$ and $\mathcal{U}$ are function spaces. In computation, functions are represented by samples on a grid, so the implemented model still manipulates finite tensors. The term *operator* describes the intended mathematical object and the architecture's treatment of spatially distributed input, not an escape from discretization.

5.2 Discretization dependence and discretization awareness

A neural operator is often designed so that its learned kernel can be evaluated on more than one resolution. That does not guarantee accurate zero-shot resolution transfer. Training data, Fourier truncation, normalization, grid spacing, and unresolved physical scales still matter. Version 1 therefore validates on the same accepted grid first.

5.3 Local versus global receptive field

A local convolution combines information from nearby grid points. Many layers are needed for distant regions to communicate. Spectral convolution transforms the field into Fourier modes and can couple information globally in one layer because each Fourier coefficient depends on the full spatial domain.

That property is attractive for MHD, where large-scale magnetic topology and localized current sheets interact. It is not a proof that spectral learning will resolve every thin structure.

5.4 Why Fourier structure is natural for this project

The synthetic solver already uses a doubly periodic Fourier pseudo-spectral representation. The domain is periodic and rectangular, with no physical wall. A Fourier neural operator therefore aligns naturally with the topology of the learning grid. This is an architectural convenience and inductive bias, not an additional reduced-MHD equation.

6 The Fourier neural operator from first principles

6.1 Version-1 architecture choice

The primary data-only neural operator is a two-dimensional Fourier neural operator (FNO) inspired by the operator-learning formulation of Li and collaborators [4]. The architecture has four conceptual stages:

1. **lifting**: map the five physical/input channels to a wider latent channel space;
2. **spectral blocks**: repeatedly combine global Fourier convolution with local channel mixing;
3. **nonlinearity**: apply a pointwise activation after each block; and
4. **projection**: map latent channels to the two target fields.

6.2 Lifting

At each grid point define the five-component input vector

$$\mathbf{a}(x, y) = [\tilde{\psi}_0, \tilde{U}_0, \tilde{\eta}, \tilde{\nu}, \tilde{t}]^T.$$

A learned pointwise affine map lifts it to width d :

$$\mathbf{v}_0(x, y) = P\mathbf{a}(x, y) + \mathbf{b}_P, \quad P \in \mathbb{R}^{d \times 5}.$$

The lifting operation does not mix different spatial locations. It creates a richer local feature vector before global spectral mixing.

6.3 One spectral layer

For layer ℓ , let $\mathbf{v}_\ell(x, y) \in \mathbb{R}^d$. A simplified FNO block is

$$\mathbf{v}_{\ell+1}(x, y) = \sigma\left(W_\ell \mathbf{v}_\ell(x, y) + \mathcal{F}^{-1}(R_\ell \cdot \mathcal{F}[\mathbf{v}_\ell])(x, y) + \mathbf{b}_\ell\right). \quad (6.1)$$

Here:

- W_ℓ is a learned pointwise channel-mixing matrix;
- $\mathcal{F}$ and $\mathcal{F}^{-1}$ are the two-dimensional discrete Fourier transform and inverse transform;
- R_ℓ contains learned complex weights acting on selected Fourier modes;
- σ is a nonlinear activation; and
- the dot indicates multiplication across input/output latent channels for each retained wave number.

6.4 Spectral convolution in component form

Let $\hat{v}_{\ell,c}(k_x, k_y)$ be Fourier coefficient (k_x, k_y) of latent channel c . For an output channel r ,

$$\hat{z}_{\ell,r}(k_x, k_y) = \sum_{c=1}^d R_{\ell,rc}(k_x, k_y) \hat{v}_{\ell,c}(k_x, k_y)$$

for retained modes. Unretained modes of the learned spectral branch are set to zero. The inverse FFT returns $z_{\ell,r}(x, y)$.

Each learned $R_{\ell,rc}(k)$ is complex because Fourier modes carry phase. Implementations with real-valued input commonly use a real FFT representation and parameterize the independent half-spectrum.

6.5 Why there is also a local W_ℓ branch

Keeping only a finite number of learned Fourier modes would discard high-frequency content inside the spectral branch. The local pointwise branch $W_\ell \mathbf{v}_\ell$ provides an additional path for information and channel transformation. The nonlinear combination of local and spectral branches is more expressive than either alone.

6.6 Mode truncation

Let the retained learned spectral band contain M_x modes in x and M_y modes in y under the implementation's FFT ordering. The choice controls a tradeoff:

- too few modes can smooth thin current sheets and miss high-wavenumber structure;
- more modes increase parameter count and memory; and
- modes near the numerical cutoff may carry discretization noise rather than trustworthy physics.

The mode count is therefore a validation-selected hyperparameter and must remain within the resolved spectral content of the accepted reference data.

6.7 Periodic domain and padding

Many FNO examples pad nonperiodic domains to reduce wrap-around artifacts. The present version-1 domain is physically periodic, so the primary baseline uses **no nonperiodic padding**. Adding arbitrary zero padding would change the represented topology and must be treated as an architecture ablation rather than a default preprocessing trick.

6.8 Activation function

The data-only FNO does not require fourth-order automatic spatial derivatives, so several smooth activations are technically possible. For continuity with Module 5 and stable high-order diagnostic behavior, version 1 uses the Gaussian error linear unit (GELU) or another explicitly frozen smooth activation selected before the final test. If Module 5 requires differentiability with respect to scalar time through the network, the chosen activation must support that derivative reliably.

For the primary matched comparison, Module 4 and Module 5 use the same activation.

6.9 Projection

After L spectral blocks, a pointwise projection maps latent features to two outputs,

$$\mathbf{q}(x, y) = Q_2 \sigma(Q_1 \mathbf{v}_L + \mathbf{b}_1) + \mathbf{b}_2, \quad \mathbf{q} = [\hat{\psi}, \hat{U}]^T.$$

The inverse field transforms then recover $(\hat{\psi}, \hat{U})$.

6.10 A frozen starting architecture

A defensible initial configuration is:

- grid: the common accepted Module 3 learning grid;
- input channels: 5;
- latent width: $d = 32$;
- spectral blocks: $L = 4$;
- retained modes: $M_x = M_y = 16$ as a starting point, validated against current-sheet fidelity;
- activation: GELU;
- output channels: 2; and
- no nonperiodic padding.

These numbers are not physical constants. The exact trainable-parameter count must be logged automatically by the implementation.

Important distinction: FNO is a baseline choice, not the definition of operator learning

DeepONet, graph neural operators, convolutional encoder-decoders, U-Nets, and other function-to-function architectures can also represent operator maps. Version 1 freezes an FNO because the current data live on a common periodic Cartesian grid. A future unstructured M3D-C1 mesh may motivate a different operator architecture.

7 Scalar conditioning: time, resistivity, and viscosity

7.1 Why scalar context is broadcast

The same η , ν , and target t apply to every spatial point in one example. Broadcasting them creates constant channels that are available to every lifted spatial location. The lifting and later nonlinear blocks can then learn interactions such as “the same initial current sheet evolves differently when η changes.”

7.2 Time-conditioned operator versus recurrent integration

The map

$$(\psi_0, U_0, \eta, \nu, t) \mapsto (\psi_t, U_t)$$

predicts the requested time directly. To obtain $t = 3$, the network need not first predict $t = 1$ and $t = 2$. This avoids recursive error accumulation, but it does not guarantee extrapolation beyond the times represented in training.

7.3 Continuous scalar input does not guarantee continuous physical accuracy

Because t , η , and ν are numerical inputs, the network can be evaluated between sampled scalar values. The resulting prediction is only an interpolation candidate. Accuracy must be checked against independent accepted solver output.

7.4 Potential alternative conditioning schemes

Alternatives include a separate scalar multilayer perceptron whose embedding modulates spectral blocks, feature-wise affine modulation, or concatenation after lifting. These may improve parameter conditioning but introduce additional architectural choices. Broadcasting is retained as the transparent primary baseline.

8 Physical nondimensionalization and neural preprocessing

8.1 Two transformations with different meanings

Module 1 nondimensionalizes the physical PDE. Module 4 may then transform already dimensionless values for optimization. These operations must remain separate.

Transformation	Purpose	Scientific consequence
Physical nondimensionalization	Express MHD using reference scales.	Determines the dimensionless equations and interpretation of η, ν, t .
Neural preprocessing	Improve numerical conditioning of inputs and targets.	Changes numbers presented to the network but is invertible and fitted only from training data.

8.2 Field standardization

The primary view uses one training-only transform for each physical channel,

$$\tilde{\psi} = \frac{\psi - m_{\psi}}{s_{\psi}}, \quad \tilde{U} = \frac{U - m_U}{s_U},$$

with $s_{\psi}, s_U > 0$. The same ψ transform applies to ψ_0 and target ψ_t ; likewise for U .

8.3 How field statistics are weighted

If statistics were computed by naively concatenating every scalar grid value from every saved time, cases with more target times could dominate. Module 3 therefore freezes a weighting policy. The primary policy is case-balanced first, target-time balanced second, and area balanced third.

8.4 Transport coefficients

For positive coefficients spanning orders of magnitude, define

$$r_\eta = \log_{10} \eta, \quad r_\nu = \log_{10} \nu$$

before affine scaling. If the range is narrow, direct affine scaling may be adequate. The transform is selected from training-design considerations and frozen before final testing.

8.5 Target time

For training interval $t \in [t_{\min}, t_{\max}]$, a simple affine scale is

$$\tilde{t} = 2 \frac{t - t_{\min}}{t_{\max} - t_{\min}} - 1.$$

Testing at $t > t_{\max}$ produces $\tilde{t} > 1$ and is genuine scalar extrapolation.

8.6 Leakage-free fitting

Every learned transform must be fitted from training cases only:

$$\mathcal{T} = \text{fitTransform}(\mathcal{D}_{\text{train}}).$$

The same stored transform is then applied unchanged to validation and test cases.

8.7 Gauge and mean conventions

The dataset fixes the spatial mean of U to zero. Arbitrary constant offsets in U would add nonphysical variation to the regression target. The mean convention must be common across all cases before training. The same principle applies to the recorded mean of ψ .

9 Partitioning and leakage control

9.1 Training, validation, and test roles

The training set updates network weights. The validation set selects architecture, mode count, width, learning rate, training duration, and checkpoints. The test set is untouched until the design is frozen.

9.2 The split unit is the complete physical case

Let $\mathcal{C}_{\text{tr}}$, $\mathcal{C}_{\text{va}}$, and $\mathcal{C}_{\text{te}}$ contain physical-case identifiers. They must be disjoint. Every target-time pair derived from one physical trajectory inherits its case partition.

Important distinction: time-snapshot leakage

Putting early times from one trajectory in training and later times from the same trajectory in test does not measure unseen-case generalization. The model has already seen the exact initial state and physical parameters of that trajectory.

9.3 Split before transforms and pair sampling

The correct order is:

1. validate accepted physical cases;
2. freeze whole-case train/validation/test assignments;
3. fit preprocessing from training cases only;
4. choose target-time pair rules within each partition; and
5. train and tune without test inspection.

9.4 Parameter interpolation versus extrapolation

In the (η, ν) plane, an unseen test point inside a declared support such as the convex hull of training coefficients may be labeled interpolation. A point outside requires extrapolation. The current A_0 family is encoded through U_0 and should be analyzed separately as initial-state amplitude generalization.

9.5 Initial-state generalization in version 1

Because

$$\psi_0 = \sin x \sin y, \quad U_0 = A_0(\cos x - \cos y),$$

all initial states lie on a very low-dimensional family. Holding out $A_0 = 0.10$ while training on $A_0 = 0.05$ and 0.20 tests amplitude interpolation of one fixed shape. It does not test a new magnetic topology or new flow morphology.

10 The strict data-only objective

10.1 Full-field pointwise errors

For example (c, n) and grid index (i, j) , define standardized errors

$$e_{\psi,c,n,i,j} = \widehat{\psi}_{c,n,i,j} - \widetilde{\psi}_{c,n,i,j},$$

$$e_{U,c,n,i,j} = \widehat{U}_{c,n,i,j} - \widetilde{U}_{c,n,i,j}.$$

The per-field area-mean squared errors are

$$E_{\psi,c,n} = \frac{1}{N_x N_y} \sum_{i,j} e_{\psi,c,n,i,j}^2, \quad E_{U,c,n} = \frac{1}{N_x N_y} \sum_{i,j} e_{U,c,n,i,j}^2.$$

10.2 Channel-balanced operator loss

After separate output standardization, the primary example loss is

$$\ell_{c,n} = \lambda_{\psi} E_{\psi,c,n} + \lambda_U E_{U,c,n}, \quad \lambda_{\psi} = \lambda_U = 1.$$

10.3 Case-balanced empirical risk

If each case contributes a different number of selected target times, a global pair average weights dense cases more heavily. The primary training risk first averages over target times within each selected case and then over cases:

$$\mathcal{L}_{\text{data}} = \frac{1}{|\mathcal{B}_C|} \sum_{c \in \mathcal{B}_C} \frac{1}{|\mathcal{B}_t(c)|} \sum_{n \in \mathcal{B}_t(c)} \ell_{c,n}. \quad (10.1)$$

10.4 The strict Module 4 loss

Frozen project convention : value-only full-field baseline

The primary Module 4 objective is

$$\mathcal{L}_{\text{train}} = \mathcal{L}_{\text{data}}.$$

No target J , target ω , PDE residual, gradient target, spectral-coefficient target, initial-condition penalty, boundary penalty, gauge penalty, or global-balance penalty may update the weights.

10.5 Are spectral losses still data-only?

A loss comparing predicted and target Fourier coefficients contains no PDE and is data-supervised in a broad sense. However, it supplies frequency-weighted information beyond the strict pointwise field loss. The same is true for gradient losses, current losses, or vorticity losses. These are useful enhanced-data baselines but must not silently replace the primary value-only experiment.

10.6 Why squared error is reasonable and limited

MSE is differentiable and strongly penalizes large errors, but area-averaging can underweight a thin current sheet because the sheet occupies little area. That limitation is handled in evaluation and controlled loss ablations rather than by changing the primary objective after seeing test results.

10.7 Weight decay and numerical regularization

Weight decay, optimizer momentum, learning-rate schedules, and gradient clipping are numerical/statistical training choices, not MHD physics. They may be used if declared and matched appropriately in Module 5.

11 Training the operator

11.1 What training changes

The operator contains trainable real and complex-valued parameters, represented internally by real tensors. The loss is a scalar function $\mathcal{L}_{\text{train}}(\theta)$. Backpropagation computes

$$\nabla_{\theta} \mathcal{L}_{\text{train}}.$$

An optimizer uses this gradient to update the parameters.

11.2 Backpropagation through FFT operations

The discrete FFT, complex multiplication, inverse FFT, pointwise affine maps, and activations are differentiable operations in modern frameworks. Backpropagation therefore propagates field-loss sensitivities through every FNO layer into its spectral and local weights.

This training derivative is with respect to network parameters. It should not be confused with physical derivatives such as $\partial_x \psi$ or $\partial_t \psi$.

11.3 Mini-batches

A batch contains complete full-field examples, not isolated pixels. For B examples on a 128×128 grid with several latent channels, activation memory can be substantial. Batch size is therefore constrained by accelerator memory and must be recorded.

11.4 Hierarchical batch sampling

A defensible sampler:

1. selects physical cases approximately uniformly;
2. selects one or more target times within each chosen case; and
3. loads the complete field pair for each selected (c, n) .

This controls correlation better than sampling uniformly from a flat list dominated by trajectories with many saved times.

11.5 Adam and learning-rate schedule

Adam [3] is a practical starting optimizer. A typical staged schedule begins with a moderate learning rate, reduces it when validation error plateaus or by a predeclared cosine schedule, and saves the checkpoint with the best declared validation metric. The exact schedule is a hyperparameter, not a physical law.

11.6 Epoch, update, and examples processed

An optimizer update changes the network once. An epoch approximately processes one pass through a declared finite pair set. If target times are resampled each epoch, the meaning of epoch becomes approximate. The project therefore logs updates, examples processed, and wall-clock time in addition to epochs.

11.7 Repeated random seeds

Initialization, batch order, and accelerator kernels can cause run-to-run variation. Final comparisons should train several seeds and report mean, spread, and failures rather than only the best run.

11.8 Staged implementation strategy

Before full training:

1. verify exact HDF5-to-tensor round trip;
2. verify $t = 0$ identity examples;
3. overfit one target pair;
4. overfit one complete trajectory with several target times;
5. train a small multi-case model;
6. inspect validation fields and spectra;
7. freeze the primary configuration; and
8. train repeated seeds for the untouched test evaluation.

Overfitting a tiny dataset is a debugging test, not evidence of generalization.

12 Hyperparameters and controlled architecture selection

12.1 Trainable parameters versus hyperparameters

Trainable parameters are updated by gradient descent. Hyperparameters include latent width, number of spectral layers, retained mode counts, activation, optimizer, learning rate, batch size, weight decay, target-time sampling, and early-stopping patience.

12.2 Mode count is physically consequential

Current density contains two spatial derivatives. For a Fourier mode of wave number k , the current contribution scales as $k^2\psi_k$. Consequently, a network that predicts low modes accurately while smoothing high modes can achieve good ψ MSE but poor J . Mode-count selection must therefore inspect derivative-sensitive validation diagnostics, not only field MSE.

12.3 Width and depth

Greater width increases the number of latent channels available at each wave number. Greater depth composes more nonlinear global-local transformations. Both increase capacity and compute. A larger model is not automatically more physical or more data-efficient.

12.4 Validation metric selection

A predeclared primary validation score may combine relative L_2 field errors for ψ and U . Secondary validation diagnostics can include J , ω , energy, and peak-current errors. If secondary diagnostics choose the checkpoint or architecture, that influence must be disclosed and applied consistently to Module 5.

12.5 Search budget fairness

One method should not receive vastly more architecture trials or manual tuning than the other. The project records the search space, number of trials, compute budget, selection rule, and final frozen configuration for both Modules 4 and 5.

13 Complete-field inference

13.1 One operator query

Given a new accepted-style initial state and context:

1. validate grid, channels, and metadata;
2. apply stored training-only transforms;
3. broadcast transformed η , ν , and t ;
4. evaluate the FNO once on the complete field;
5. inverse-transform the two output channels; and
6. return $(\hat{\psi}_t, \hat{U}_t)$ with model and preprocessing provenance.

13.2 A trajectory

To reconstruct a trajectory at times $t_1, \dots, t_m$, evaluate the time-conditioned operator for each requested t_k . These queries can be batched because they share the same initial fields but differ in the time channel.

13.3 Inference cost must match the scientific task

The relevant comparison is full-field prediction time, not the cost of one pixel or one Fourier coefficient. Report end-to-end latency including tensor transforms and device transfer when comparing with the reference solver.

13.4 No recursive error accumulation, but possible temporal inconsistency

Direct time conditioning avoids recursive propagation of the model's own previous output. However, predictions at t_1 and t_2 are produced independently and need not form a dynamically consistent trajectory. Module 5 physics terms and evaluation-only temporal diagnostics can expose such inconsistency.

14 Evaluation on complete unseen fields

14.1 Why training loss is not the scientific result

Training loss measures fit in transformed coordinates on observed pairs. Scientific evaluation uses inverse-transformed dimensionless fields from whole held-out cases and target times.

14.2 Absolute and relative field errors

For $a \in \{\psi, U\}$,

$$e_a(x_i, y_j) = \hat{a}_{ij} - a_{ij}.$$

A discrete relative L_2 error is

$$\text{RelL2}(a) = \frac{\left(\sum_{i,j} |\hat{a}_{ij} - a_{ij}|^2\right)^{1/2}}{\max\left[\left(\sum_{i,j} |a_{ij}|^2\right)^{1/2}, \epsilon\right]}.$$

Also report absolute RMSE and relative maximum error.

14.3 Per-case and aggregate reporting

Report each held-out physical case separately before aggregate statistics. A mean over all target-time pairs can hide one failed parameter regime. Recommended summaries include median, mean, worst case, and distribution across random seeds.

14.4 Time-resolved error

For a held-out trajectory, plot or tabulate field error versus target time. Reconnection is a transient event; an average across time can hide a severe error near current-sheet formation.

14.5 Spectral error

On the common periodic grid, compare Fourier spectra of predicted and reference fields. Define shell or mode-wise energy errors and inspect whether the operator loses high-wavenumber content as the current sheet sharpens.

14.6 Amplitude error versus phase/topology error

A prediction can have similar energy but place an island or X-point at the wrong location. Contours remain useful visualization tools for evaluation, but metrics must quantify topology and critical-point location rather than relying on visual similarity alone.

15 Physical diagnostics that remain evaluation-only

15.1 Derived current and vorticity

For predicted complete fields, use the same declared spectral derivative convention as the reference grid:

$$\hat{J} = -\nabla_{\perp}^2 \hat{\psi}, \quad \hat{\omega} = \nabla_{\perp}^2 \hat{U}.$$

This is preferable to image processing or arbitrary finite differences because it matches the periodic numerical representation.

15.2 Magnetic and kinetic energy

Evaluate

$$\hat{E}_B = \frac{1}{2} \int_{\Omega} |\nabla \hat{\psi}|^2 dA, \quad \hat{E}_K = \frac{1}{2} \int_{\Omega} |\nabla \hat{U}|^2 dA.$$

Compare both instantaneous values and histories with the reference trajectory.

15.3 Peak current and current-sheet structure

Record

$$\hat{J}_{\max}(t) = \|\hat{J}\|_{\infty},$$

its location, sheet width, and if available the same orientation/width diagnostic used by Module 2. Thin sheets are precisely where field MSE can be misleading.

15.4 O-points, X-points, and reconnected flux

Apply the Module 2 critical-point and reconnection definitions to the predicted $\hat{\psi}$. Compare O/X locations, O-to-X flux contrast, reconnected flux, and event timing. These diagnostics test whether the surrogate reproduces magnetic topology rather than only array averages.

15.5 Periodicity

An FNO on a periodic FFT grid naturally uses wrap-around Fourier representation, but implementation mistakes can still enter through preprocessing or tensor conversion. Evaluate periodic seam and derivative consistency explicitly when reconstructing continuous diagnostics.

15.6 Evaluation-only reduced-MHD residuals

The predicted fields can be inserted into

$$\mathcal{R}_\psi = \psi_t + [U, \psi] - \eta \nabla_\perp^2 \psi,$$

$$\mathcal{R}_\omega = \omega_t + [U, \omega] - [J, \psi] - \nu \nabla_\perp^2 \omega.$$

For a time-conditioned operator, spatial derivatives can be computed spectrally and ∂_t may be obtained by automatic differentiation with respect to the scalar time channel if the implementation preserves that graph, or by a carefully defined temporal diagnostic. These residuals remain evaluation-only in Module 4.

Important distinction: diagnostic use versus training use

Computing a PDE residual after training does not turn the model into a PINN. Using that residual to update weights does. If residual-based model selection is used, it influences the experimental design and must be disclosed even though it is not a gradient term.

15.7 Energy-balance defect

For the predicted trajectory evaluate the reduced-MHD dissipation identity

$$\frac{dE}{dt} = -\eta \int J^2 dA - \nu \int \omega^2 dA$$

as an evaluation diagnostic. A data-only operator is not required to satisfy it exactly.

16 Generalization experiments

16.1 Transport-parameter interpolation

Hold out entire (η, ν) cases inside the training support. The initial field may still resemble training states, but the model must infer dynamics at a new transport combination.

16.2 Transport-parameter extrapolation

Evaluate coefficients outside the training support. This is more difficult because the operator must extrapolate the effect of diffusion rather than interpolate it.

16.3 Time interpolation

If target times inside $[0, T]$ are sparsely labeled for training, evaluate unlabeled intermediate times from held-out cases. This tests continuous time conditioning within the represented interval.

16.4 Time extrapolation

Train only through T_{train} and evaluate at $t > T_{\text{train}}$. This is a separate experiment. A continuous time input does not imply reliable forecasting.

16.5 Initial-amplitude interpolation

Under the current family, holding out one A_0 tests whether the operator can infer an intermediate complete U_0 amplitude. The operator receives the field itself, not the label A_0 .

16.6 Initial-shape generalization

A future stronger experiment should generate additional verified ψ_0 and/or U_0 shapes. Only then can the project test whether the learned operator generalizes between distinct initial spatial structures. This extension belongs first in Module 2 case generation and Module 3 schema/view design.

16.7 Resolution transfer

An FNO can in principle be evaluated on a different discretization when the spectral implementation and learned modes are compatible. Nevertheless, the current baseline should first test same-grid accuracy. A resolution-transfer experiment must use an independent higher-resolution reference and report whether thin current sheets remain resolved.

17 Data efficiency and learning curves

17.1 Several distinct data budgets

For a full-field operator, “number of training points” is no longer the most meaningful budget. Report separately:

- number of independent physical trajectories;
- number of selected target times per trajectory;
- spatial resolution of each field;
- number of observed output channels; and
- total reference-solver cost required to generate the labeled fields.

17.2 Trajectory scarcity versus snapshot scarcity

Reducing target times from 100 to 20 within the same 27 trajectories is different from reducing the number of trajectories from 27 to 6. The first removes temporal labels but keeps physical-case diversity. The second removes independent variation in initial conditions and transport parameters.

17.3 Learning curves

A useful study varies both axes:

$$N_{\text{case}} \in \{\text{small, medium, full}\}, \quad N_{t,\text{label}} \in \{\text{sparse, medium, dense}\}.$$

Module 4 and Module 5 should use identical labeled pairs at each budget.

17.4 Reference cost matters

One full-field target may contain many scalar values, but its expensive information source is the reference simulation. Data-efficiency claims should therefore report solver runs and saved fields, not merely scalar tensor elements.

18 Failure modes and their signatures

18.1 Underfitting

Training and validation errors both remain high; broad topology may be reproduced while small structures are absent. Increase capacity or improve optimization only after confirming data and transforms.

18.2 Overfitting

Training error falls while unseen-case error worsens. Large spectral models can memorize a small set of trajectories despite many apparent target-time pairs.

18.3 Trajectory memorization

If case diversity is small, the network may interpolate labels associated with familiar initial fields rather than learn a robust solution operator. Dense temporal sampling cannot replace independent initial-state diversity.

18.4 High-frequency smoothing

Predicted ψ contours look good but J is too small or too broad. The spectral tail of $\hat{\psi}$ decays too rapidly compared with the reference. Increase validated mode capacity or reconsider the architecture; do not infer accuracy from contours alone.

18.5 Spectral ringing

Sharp structures can produce oscillatory errors around a current sheet. Inspect physical-space error and spectra. More modes are not always the solution if the reference itself is near its resolution limit.

18.6 Scalar-conditioning failure

The network reproduces average behavior but responds weakly to changes in η or ν . Compare cases with identical or similar initial fields across transport coefficients and inspect sensitivity of outputs to context channels.

18.7 Temporal inconsistency

Predictions at neighboring target times are individually plausible but jump nonphysically because each query is independent. Time-resolved diagnostics and evaluation-only PDE/energy defects reveal the problem.

18.8 Gauge contamination

An arbitrary spatially constant offset appears in U . This can give small velocity error but large direct U loss. Enforce the dataset gauge consistently rather than post-hoc shifting test predictions without declaring the procedure.

18.9 Grid transposition

Square fields can be accidentally transposed and still look plausible. Module 3 requires explicit axis metadata and non-square sentinel tests for packing parity.

18.10 Rendered-image contamination

If a contour image or color map enters the learning pipeline, numerical meaning depends on plotting choices and precision is lost. The remedy is to return to raw HDF5 field arrays.

18.11 Test leakage

Using test spectra to select mode count, test field means to normalize outputs, or target times from the same physical trajectory across partitions produces optimistic results.

18.12 False M3D-C1 interpretation

A network can be excellent on the synthetic periodic two-field system and still have no demonstrated accuracy on M3D-C1. Architecture similarity does not close the model-discrepancy gap.

19 Fair comparison with the Module 5 physics-informed operator

19.1 What should be held fixed

The primary comparison should hold fixed:

- physical-case train/validation/test split;
- labeled operator pairs;
- input/target channel semantics;
- grid and tensor packing;
- preprocessing transforms;

- FNO architecture, width, layers, modes, activation, and projection unless a controlled reason requires otherwise;
- data-loss definition;
- random seeds where meaningful; and
- scientific evaluation metrics.

Module 5 additionally receives physics collocation/evaluation information and physics-loss terms.

19.2 Three notions of fairness

Label fairness means the same reference field labels. **Architecture fairness** means comparable representational capacity. **Compute fairness** means runtime and memory differences are reported rather than hidden. Physics-informed training may use much more compute while using the same labels.

19.3 Matched labeled-data budgets

For each budget, the immutable set of labeled (c, n) pairs is identical in Modules 4 and 5. Additional Module 5 collocation times or physics evaluations do not count as new labels but must be counted in compute.

19.4 Ablation ladder

A useful sequence is:

1. data-only FNO;
2. data-only FNO plus optional enhanced data loss, if studied separately;
3. data + magnetic-flux PDE physics;
4. data + both PDEs;
5. data + PDEs + initial/gauge constraints where needed;
6. optional global balances.

This identifies which information channel produces any observed benefit.

19.5 What counts as evidence that physics helped?

Examples include lower unseen-case field and derivative error at the same labels, better current-sheet and reconnection fidelity, lower data requirement for a fixed accuracy target, or improved extrapolation without unacceptable compute or failure-rate cost. Lower training residual by itself is not sufficient.

20 Recommended version-1 baseline specification

Frozen project convention: primary data-only full-field operator

- Task: $(\psi_0, U_0, \eta, \nu, t) \mapsto (\hat{\psi}_t, \hat{U}_t)$.
- Data: accepted Module 3 full-field numerical arrays only; no rendered images.
- Split: whole physical cases before transforms and target-time pair extraction.
- Grid: one common accepted periodic production grid per physical case.
- Input channels: standardized ψ_0, U_0 plus broadcast transformed η, ν, t ; no duplicate A_0 channel.
- Target channels: standardized ψ_t, U_t .
- Architecture: 2D FNO, width 32, four spectral blocks, starting retained modes 16×16 , no nonperiodic padding, smooth frozen activation.
- Loss: case-balanced, target-time-balanced, channel-balanced full-field value MSE only.
- Optimizer: Adam with a predeclared learning-rate schedule.
- Model selection: validation cases only.
- Final reporting: repeated seeds and complete-field physical diagnostics on untouched test cases.

20.1 Why this is only a starting point

The architecture is chosen for transparency and compatibility with the periodic structured grid. A competent validation study can change width, modes, activation, or number of layers before final testing. The frozen primary comparison must then use the selected configuration consistently in Module 5.

20.2 Suggested controlled ablations

Useful ablations include:

- 8, 16, and 24 retained modes;
- width 16, 32, and 64;
- scalar broadcast versus learned scalar embedding;
- direct time-conditioned operator versus autoregressive time-stepper;
- value-only versus value-plus-spectral data loss;
- with and without deterministic coordinate channels; and
- FNO versus a conventional U-Net of comparable capacity.

Each ablation answers a different architectural question and should not be mixed into the physics-informed comparison without labeling it.

21 Worked miniature examples

21.1 Example 1: pure resistive decay

With $U_0 = 0$ and the initial magnetic eigenmode from Module 2,

$$\psi(x, y, t) = e^{-2\eta t} \sin x \sin y, \quad U = 0.$$

For this restricted family the exact solution operator is

$$(\sin x \sin y, 0, \eta, \nu, t) \mapsto (e^{-2\eta t} \sin x \sin y, 0).$$

A network can learn the scalar decay law while preserving the spatial mode. This is a useful debugging problem because truth is analytic.

21.2 Example 2: why full-field input contains A_0

Consider two initial flow amplitudes,

$$U_0^{(1)} = 0.05(\cos x - \cos y), \quad U_0^{(2)} = 0.20(\cos x - \cos y).$$

The input arrays differ point by point. The network does not need a separate label to know which amplitude it received. Nevertheless, both fields share exactly the same normalized shape, so this experiment does not represent arbitrary initial-flow learning.

21.3 Example 3: field MSE can hide current error

Let a predicted magnetic flux contain a small high-frequency error,

$$\hat{\psi} = \psi + \epsilon \sin(kx) \sin(ky).$$

The flux error amplitude is $O(\epsilon)$, but

$$\hat{J} - J = -\nabla_{\perp}^2 (\hat{\psi} - \psi) = 2k^2 \epsilon \sin(kx) \sin(ky),$$

so current error grows like k^2 . This is why retained FNO modes and derivative diagnostics matter.

21.4 Example 4: pair count is not case count

Suppose 9 training trajectories each contribute 80 target times. The training loader sees 720 operator pairs, but the model has only 9 independent initial-value problems. A validation set containing another time from those same 9 trajectories would not test new-case generalization.

22 Reproducibility and implementation handoff

22.1 Configuration that must be saved

Each run records at least:

- dataset release and operator-view identifiers;
- immutable split manifest digest;
- selected target-time rule and labeled pair IDs;
- grid signature and channel order;
- field and scalar preprocessing parameters;
- FNO width, layer count, retained mode counts, activation, and padding rule;
- exact parameter count;
- optimizer, learning-rate schedule, weight decay, batch size, and gradient clipping;
- random seed and framework/device information;
- checkpoint selection metric and patience;
- training updates, examples processed, wall-clock time, and peak memory; and
- source revision/configuration digest.

22.2 Acceptance tests before full training

Mandatory tests include:

1. exact HDF5-to-tensor channel and axis round trip;
2. non-square sentinel grid packing test;
3. $t = 0$ identity target construction;
4. inverse preprocessing round trip;
5. one-batch forward shape and finite-value check;
6. one-batch backward finite-gradient check;
7. overfit-one-pair test;
8. overfit-one-trajectory test;
9. periodic FFT ordering test for predicted diagnostic derivatives; and
10. deterministic evaluation-mode repeatability.

22.3 Minimum run products

Save the best checkpoint, final configuration, training/validation histories, seed information, test predictions or reproducible prediction indices, per-case metrics, time-resolved metrics, field/spectral error products, J/ω diagnostics, topology/reconnection metrics, runtime data, and a machine-readable summary.

22.4 Runtime quantities

Report training wall time, accelerator type, peak memory, full-field inference latency, throughput for batched target times, parameter count, and reference-solver cost. A model that saves labels but requires dramatically more training compute may still be useful, but the tradeoff must be visible.

23 Common misconceptions

23.1 “A neural operator takes images”

It takes numerical fields. A 2D tensor is not automatically an image; each entry has a physical scalar meaning tied to coordinates, normalization, and provenance.

23.2 “More target times mean more independent simulations”

They produce more supervised pairs, not more independent trajectories.

23.3 “The full initial field means arbitrary initial-condition generalization”

Not in version 1. The initial fields still come from a narrow analytic family.

23.4 “ A_0 disappeared from the physics”

No. It still defines the version-1 initial flow and is stored as metadata. It is simply encoded directly in U_0 and therefore need not be duplicated as a neural input channel.

23.5 “An FNO is physics-informed because it uses Fourier modes”

No. Fourier transformation is an architectural/numerical representation. The model becomes physics-informed only when declared physical information affects training.

23.6 “Low field MSE guarantees good reconnection physics”

No. Current sheets, derivatives, X-points, and event timing can be wrong despite visually good fields.

23.7 “A time-conditioned operator forecasts automatically”

No. Evaluation within the training time interval is interpolation. Forecasting requires explicit time extrapolation.

23.8 “More Fourier modes always improve the model”

No. More modes increase capacity and cost and may fit numerical noise. Validation must determine whether they improve resolved physical diagnostics.

23.9 “FNO resolution independence means exact super-resolution”

No. An architecture may accept another grid, but accuracy depends on training support and physical resolution.

23.10 “Broadcasting η means resistivity varies in space”

No. The broadcast channel is constant over the field for one case. It is only an input representation.

23.11 “A good synthetic operator is already an M3D-C1 surrogate”

No. Genuine M3D-C1 field mappings, geometry, closures, meshes, and validation are still required.

24 Module 4 computational contract

24.1 Input-output contract

For one operator example,

$$(\psi_0[N_x, N_y], U_0[N_x, N_y], \eta, \nu, t) \mapsto (\hat{\psi}_t[N_x, N_y], \hat{U}_t[N_x, N_y]).$$

The fields are raw numerical scalar arrays on the common periodic grid. A_0 remains metadata and is already encoded in U_0 for version 1.

24.2 Data contract

Only accepted Module 3 cases may enter. Whole physical cases are split before pair extraction and preprocessing. Training transforms are fitted from training cases only. No rendered image enters the learning path.

24.3 Architecture contract

The primary model is a 2D FNO with explicit lifting, four global-local spectral blocks, and two-channel projection. Exact width, modes, activation, and optimizer choices are frozen after validation and logged.

24.4 Loss contract

$$\mathcal{L}_{\text{train}} = \mathcal{L}_{\text{data}}$$

with case-balanced, target-time-balanced, channel-balanced full-field value MSE. Any derivative, Fourier-target, PDE, boundary, gauge, or balance term defines a different experiment.

24.5 Evaluation contract

Evaluate inverse-transformed complete fields on untouched cases. Report ψ, U, J, ω , spectra, energies, current-sheet measures, topology/reconnection diagnostics, evaluation-only residuals, and runtime.

24.6 Fairness contract for Module 5

Module 5 inherits the same operator task, split, labeled pairs, preprocessing, primary architecture, and data loss. Added physics information, collocation strategy, loss weighting, and added compute are separately recorded.

25 Summary and bridge to Module 5

Module 4 now learns the data-only full-field solution operator

$$(\psi_0, U_0, \eta, \nu, t) \mapsto (\hat{\psi}_t, \hat{U}_t).$$

The input and output are complete numerical scalar fields, not images. The version-1 initial perturbation amplitude A_0 remains physical metadata but is already encoded by $U_0 = A_0(\cos x - \cos y)$, so it is not duplicated as a primary input channel.

The primary architecture is a two-dimensional Fourier neural operator on the common periodic grid. A lifting layer maps the two initial fields and three broadcast scalar-context channels into a latent representation. Repeated FNO blocks combine learned low-mode spectral convolution with local pointwise channel mixing and nonlinear activation. A projection layer returns ψ and U .

The strict baseline is supervised only. Whole physical trajectories are partitioned before target-time extraction. Preprocessing is fitted from training cases only. The training objective contains only complete-field value error, averaged so that output channels, target times, and physical cases receive controlled weight.

Scientific evaluation goes well beyond tensor MSE. Current and vorticity, spectra, energies, current-sheet structure, critical points, reconnected flux, event timing, periodicity, and reduced-MHD residuals are evaluated after training. These diagnostics can reveal derivative or topology failure hidden by smooth contour plots.

The full-field representation creates a more natural bridge to future simulation exports, but the current initial-condition family remains narrow and all data remain synthetic reduced-MHD trajectories. No M3D-C1 surrogate claim follows yet.

Module 5 keeps this exact operator task and labeled-data baseline, then asks the controlled question: does adding the reduced-MHD equations and related physical constraints improve accuracy, data efficiency, extrapolation, or physical consistency enough to justify the added training complexity?

A Tensor and linear-algebra foundations

A.1 Scalar, vector, matrix, and tensor

A scalar is one number. A vector is an ordered list of numbers. A matrix is a two-index array. A tensor is a multidimensional array. In this module, “tensor” primarily means the computational array used by a machine-learning framework; it need not imply a tensor transformation law from differential geometry.

A single target field has shape $N_x \times N_y$. Two output channels have shape $N_x \times N_y \times 2$. A batch adds another axis.

A.2 Channel mixing as matrix multiplication

At one grid point, the lifting layer maps five channels to d channels:

$$\mathbf{v} = P\mathbf{a} + \mathbf{b}.$$

The same P is applied independently at every spatial point. This is equivalent to a 1×1 convolution in image-processing terminology, but here the channels are physical/latent fields rather than colors.

A.3 Complex spectral weights

A complex number is $z = a + ib$. Multiplying a Fourier coefficient by a complex weight can change both amplitude and phase. Machine-learning frameworks store these weights directly as complex tensors or as paired real and imaginary parts.

B Discrete Fourier representation used by the FNO

B.1 Periodic Fourier modes

For $L_x = L_y = 2\pi$,

$$f(x, y) = \sum_{m, n} \hat{f}_{mn} e^{i(mx + ny)}.$$

The discrete FFT maps sampled field values to discrete coefficients. The inverse FFT reconstructs the grid field under the library’s normalization convention.

B.2 Why spectral convolution is global

A Fourier coefficient

$$\hat{f}_{mn} = \sum_{j,\ell} f_{j\ell} e^{-i(mx_j + ny_\ell)}$$

depends on all grid points. Therefore modifying one retained Fourier mode changes the reconstructed field across the entire domain.

B.3 Spectral mode truncation versus solver dealiasing

The FNO's learned-mode truncation and the reference solver's two-thirds dealiasing serve different purposes. Solver dealiasing prevents nonlinear numerical products from folding unresolved modes into resolved modes. FNO mode truncation limits the learned spectral kernel for architectural efficiency. They must not be conflated.

B.4 Derivative amplification

For a mode $e^{i(k_x x + k_y y)}$,

$$\nabla_{\perp}^2 e^{i(k_x x + k_y y)} = -(k_x^2 + k_y^2) e^{i(k_x x + k_y y)}.$$

Thus high-wave-number flux error is strongly amplified in current. This mathematical fact motivates derivative-sensitive evaluation.

C Metric catalog

C.1 Mean absolute error

For $N = N_x N_y$ scalar values,

$$\text{MAE} = \frac{1}{N} \sum_{p=1}^N |\hat{a}_p - a_p|.$$

C.2 Mean squared error and root mean squared error

$$\text{MSE} = \frac{1}{N} \sum_{p=1}^N (\hat{a}_p - a_p)^2, \quad \text{RMSE} = \sqrt{\text{MSE}}.$$

C.3 Relative L_2 error

$$\text{RelL2}(a) = \frac{\|\hat{a} - a\|_2}{\max(\|a\|_2, \epsilon)}.$$

C.4 Relative maximum error

$$e_{\infty,\text{rel}}(a) = \frac{\|\hat{a} - a\|_{\infty}}{\max(\|a\|_{\infty}, \epsilon)}.$$

C.5 Spectral relative error

For Fourier coefficients,

$$e_{\text{spec}}(a) = \frac{\|\hat{\hat{a}} - \hat{a}_{\text{ref}}\|_2}{\max(\|\hat{a}_{\text{ref}}\|_2, \epsilon)},$$

where the outer hat denotes Fourier transform only in this appendix; implementation notation should avoid ambiguity with neural prediction hats.

C.6 Event-time error

If t_*^{ref} is the time of a reference peak and t_*^{pred} the predicted peak,

$$e_t = |t_*^{\text{pred}} - t_*^{\text{ref}}|.$$

Peak amplitude and peak time should be reported separately.

D Symbol table

Symbol	Meaning	Role
x, y	Periodic Cartesian coordinates	dimensionless
t	Requested target time	scalar operator context
η	Magnetic diffusivity	scalar operator context
ν	Kinematic viscosity	scalar operator context
A_0	Initial flow perturbation amplitude	metadata encoded in U_0
ψ_0, U_0	Complete initial magnetic-flux and stream-function fields	primary input fields
ψ_t, U_t	Complete reference target fields at time t	primary targets
$\hat{\psi}, \hat{U}$	Neural-operator predictions	model outputs
J	$-\nabla_{\perp}^2 \psi$	derived current
ω	$\nabla_{\perp}^2 U$	derived vorticity

Symbol	Meaning	Role
$\mathcal{S}_t$	Reduced-MHD solution operator	conceptual truth map
$\mathcal{G}_\theta$	Learned neural operator	approximation
θ	Trainable network parameters	learned
B	Batch size	training choice
C_{in}	Number of input tensor channels	5 in baseline
d	Latent channel width	32 starting value
L	Number of FNO spectral blocks	4 starting value
M_x, M_y	Learned retained Fourier-mode counts	hyperparameters
P	Lifting matrix	learned
W_ℓ	Pointwise channel-mixing matrix in layer ℓ	learned
$R_\ell(k)$	Complex learned spectral weights	learned
$\mathcal{F}, \mathcal{F}^{-1}$	Discrete Fourier transform and inverse	architecture operations
σ	Activation function	frozen hyperparameter
$\mathcal{L}_{\text{data}}$	Full-field supervised data loss	training objective
$\mathcal{L}_{\text{train}}$	Total Module 4 loss	equals $\mathcal{L}_{\text{data}}$
N_{case}	Number of independent physical trajectories	data budget
$N_{t,\text{label}}$	Selected labeled target times per trajectory	data budget

E Acronym and terminology table

Term	Meaning
AD	Automatic differentiation
FNO	Fourier neural operator
FFT	Fast Fourier transform
HDF5	Hierarchical Data Format version 5
I/O	Input/output
MHD	Magnetohydrodynamics
MSE	Mean squared error
PINN	Physics-informed neural network
RMSE	Root mean squared error
Full-field operator example	Complete initial numerical fields plus scalar context paired with complete target numerical fields

Term	Meaning
Neural operator	Learned map intended to act between functions or discretized fields
Lifting	Pointwise map from physical input channels to latent channels
Projection	Pointwise map from latent channels to output physical channels
Spectral convolution	Learned channel mixing of Fourier coefficients followed by inverse transformation
Retained modes	Fourier modes on which the FNO spectral branch has learned weights
Broadcast channel	Spatially constant field used to present one scalar context value at every grid point
Time-conditioned operator	Operator that receives the requested target time as an input and predicts that state directly
Autoregressive time-stepper	Model that repeatedly predicts a next state from a previous state
Case-balanced loss	Risk that gives controlled weight to independent physical trajectories rather than their raw pair counts
Data-only	Training uses labeled numerical observations but no physical equation or constraint term
Operator pair	One initial-state/context input paired with one target-time full-field state
Resolution transfer	Evaluation of a learned operator on a spatial discretization different from the primary training grid
M3D-C1-shaped	Workflow/interface organization intended to ease later higher-fidelity field ingestion; not a claim of physical equivalence

F Concept questions and answer sketches

Questions

- Q1.** Why is a data-only baseline necessary before adding physics-informed losses?
- Q2.** What is the exact version-1 Module 4 operator map?
- Q3.** Why are the 2D field arrays not treated as ordinary images?
- Q4.** Why does one physical trajectory create many operator pairs but only one independent case?
- Q5.** Why must the train/validation/test split occur before target-time pair extraction?
- Q6.** Why is A_0 not duplicated as a primary network input?
- Q7.** Why must A_0 still remain metadata?

- Q8.** What is the difference between a coordinate network and a neural operator?
- Q9.** What is the difference between the time-conditioned operator and an autoregressive time-stepper?
- Q10.** What does the FNO lifting layer do?
- Q11.** What does a spectral convolution layer do?
- Q12.** Why is the spectral convolution global in space?
- Q13.** Why retain a local pointwise branch in an FNO block?
- Q14.** Why can too few retained modes damage current prediction?
- Q15.** Why is zero padding not part of the primary periodic-domain baseline?
- Q16.** What do the broadcast η , ν , and t channels mean physically?
- Q17.** Why does continuous time input not prove forecasting ability?
- Q18.** What is the strict Module 4 training loss?
- Q19.** Can a Fourier-target loss still be called data-only?
- Q20.** Why is field MSE insufficient for magnetic reconnection assessment?
- Q21.** How should J and ω be derived from predicted fields?
- Q22.** Can PDE residuals be evaluated without making Module 4 a PINN?
- Q23.** What does holding out one A_0 value test in the current dataset?
- Q24.** What additional dataset change is needed to test arbitrary initial-condition shapes?
- Q25.** What should be held fixed in the Module 4 versus Module 5 comparison?
- Q26.** What claim about M3D-C1 is justified by a successful Module 4 result?

Answer sketches

- A1.** It measures what labeled field data alone can accomplish, so any change from added physics can be identified rather than assumed.
- A2.** $(\psi_0, U_0, \eta, \nu, t) \mapsto (\hat{\psi}_t, \hat{U}_t)$ on complete numerical fields.
- A3.** Each array entry is a quantitative scalar field value tied to physical coordinates and normalization; rendering introduces color maps, rasterization, and information loss.

- A4.** Target times share the same initial-value problem and are strongly correlated; pair count is not trajectory diversity.
- A5.** Otherwise different times from the same trajectory can leak across partitions and preprocessing can use held-out information.
- A6.** In version 1, $U_0 = A_0(\cos x - \cos y)$ already contains the amplitude everywhere on the grid.
- A7.** It is part of the physical case definition and provides a consistency/provenance check even if it is not duplicated in the neural tensor.
- A8.** A coordinate network returns field values at queried coordinates; a neural operator maps an entire input field/function to an entire output field/function.
- A9.** The time-conditioned operator predicts a requested target time directly from the initial state; a time-stepper recursively advances previous states.
- A10.** It converts the physical input channels at each grid point into a wider latent feature vector.
- A11.** It FFTs latent fields, applies learned complex channel mixing to retained modes, and inverse transforms the result.
- A12.** Each Fourier coefficient depends on the whole spatial domain, so changing it affects the reconstructed field globally.
- A13.** It supplies local channel transformation and an information path not limited to the learned spectral band.
- A14.** Current involves a Laplacian, so high-wavenumber flux errors are amplified by k^2 .
- A15.** The physical domain is already periodic; arbitrary zero padding changes the topology and is unnecessary for the primary representation.
- A16.** They are constant context values repeated spatially for architectural convenience; they are not spatially varying coefficients.
- A17.** The network can be evaluated at any numerical time, but accuracy outside the represented interval is extrapolation and must be tested.
- A18.** A case-balanced, time-balanced, channel-balanced complete-field value MSE and nothing else.
- A19.** Yes in a broad sense, because it uses target data rather than a PDE, but it is an enhanced-data baseline and should be labeled separately from the strict value-only baseline.
- A20.** Reconnection depends on derivatives, thin current sheets, topology, and event timing that can be wrong despite low area-averaged field error.
- A21.** Use the same periodic spectral derivative convention: $J = -\nabla_{\perp}^2 \psi$ and $\omega = \nabla_{\perp}^2 U$.

- A22.** Yes, if residuals are evaluation-only and never update the weights.
- A23.** It tests interpolation or extrapolation in the amplitude of one fixed analytic flow shape, not a new initial morphology.
- A24.** Module 2 must generate additional independently verified initial-field shapes and Module 3 must include them in the operator release/splits.
- A25.** The physical splits, labeled pairs, preprocessing, operator task, baseline architecture, data loss, seeds where meaningful, and scientific metrics.
- A26.** Only that a data-only full-field operator approximates the synthetic two-field reduced-MHD dataset over the tested domain. Genuine M3D-C1 fidelity remains untested.

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